

Module 4

45 Minutes

Smart and Connected Mobility

What do you observe?

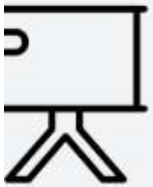
The vehicle seamlessly communicates with the passengers catering to their basic needs of work and entertainment. Alongside, it also communicates with other vehicles to ensure safe navigation on the roads.



Unit 4.1: Smarter Roads and Safer Journeys

Key Learning Outcomes

At the end of this unit, you should be able to:



- Explain the concept of intelligent transportation systems (ITS).
- Discuss the concept of connecting for safety in smart city infrastructure.
- Evaluate case studies of successful ITS implementations.

Smarter Roads and Safer Journeys

Welcome to the future of travel, where roads talk to cars, traffic lights adapt to real-time flow, and accidents become increasingly rare.

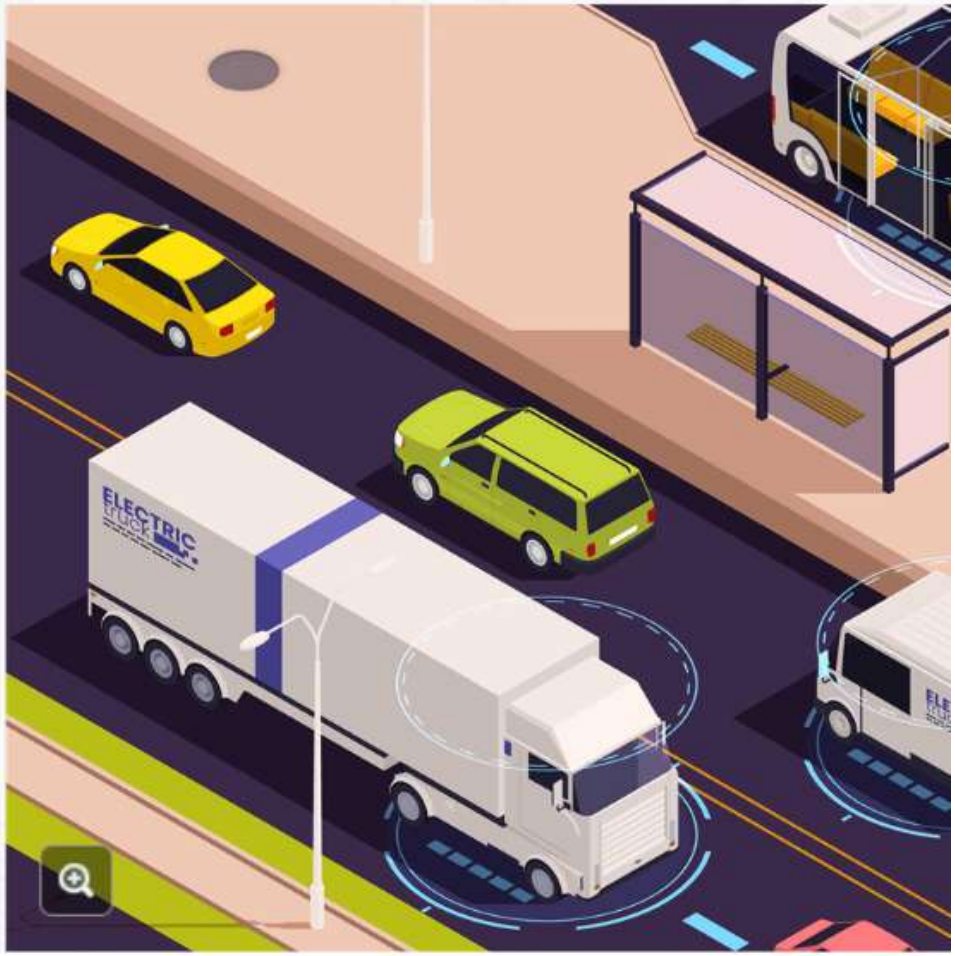
This unit explores Intelligent Transportation Systems (ITS) and how connectivity is revolutionising safety on the roads.



4.1.1: Intelligent Transportation Systems (ITS)

ITS is a broad umbrella term surrounding various technologies and systems to improve transportation networks' efficiency, safety, and sustainability.

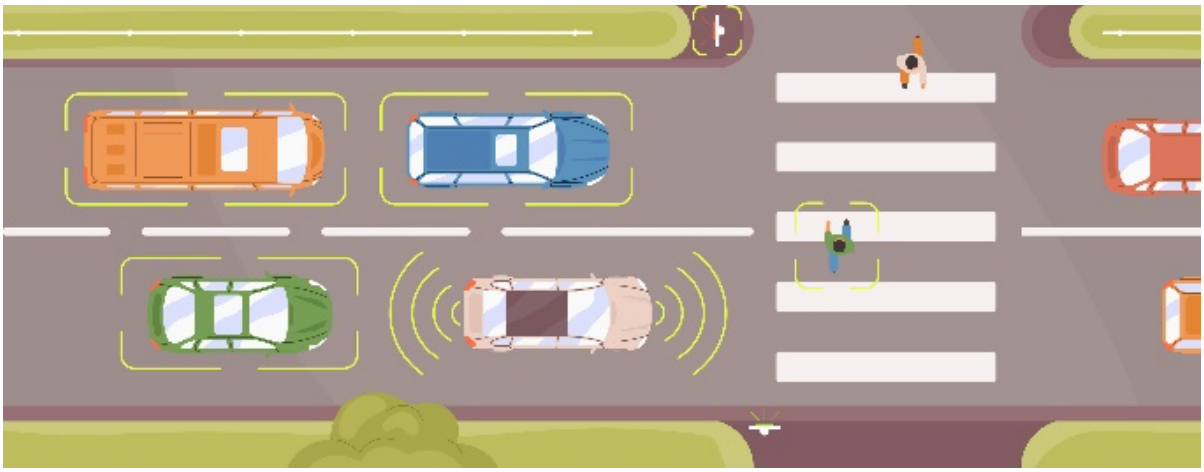
Key Pillars of ITS



Key Pillars of ITS

Below are the five main key pillars of ITS:

- Traffic Management System
- Travel Information System
- Safety System
- Public Transportation System
- Commercial Vehicle Operation



Let us now go through each one of them :

1. **Traffic Management System (TMS)**

   TMS leverages sensors, cameras, and data analysis to manage traffic flow and reduce congestion.



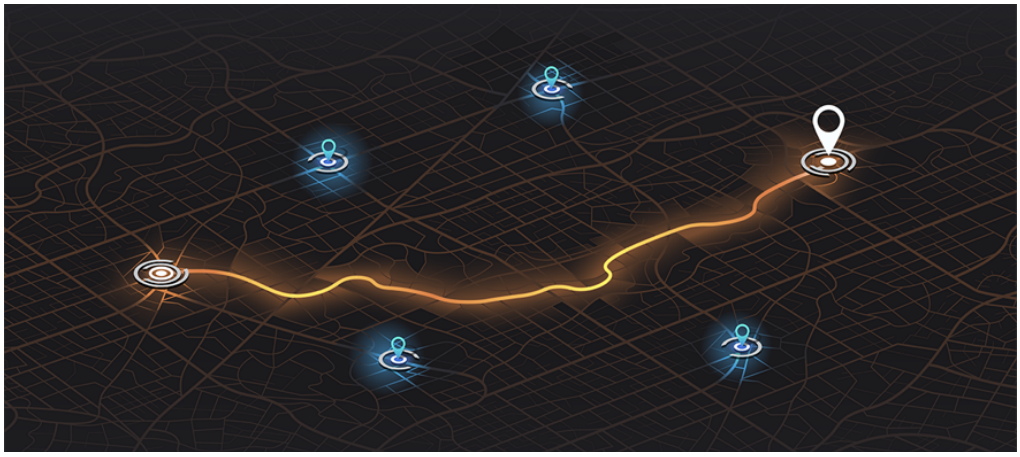
It includes:

Real-time traffic updates: Provide drivers with immediate information about congestion, accidents, and road closures.

Variable speed limits: Automatically adjust speed limits based on traffic flow to optimize traffic flow and improve safety.

Dynamic lane controls: Convert lanes in real-time to accommodate changing traffic demands.

Smart parking systems: Use sensors and software to guide drivers to available parking spaces and optimise parking utilization.

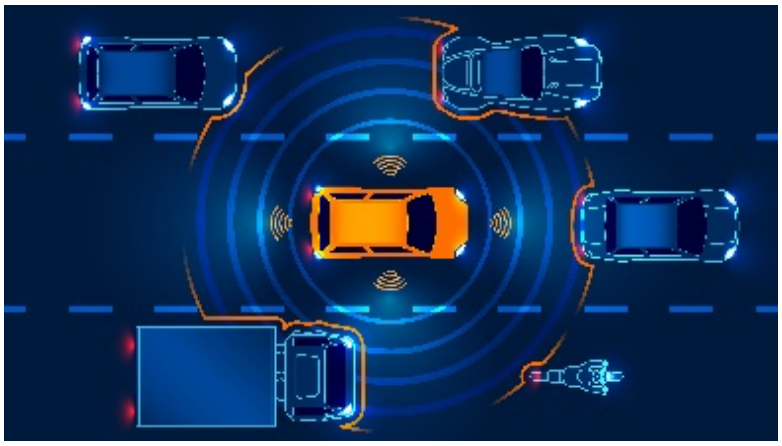


2. **Travel Information System (TIS)**

TIS provides drivers with crucial information to make informed decisions and navigate seamlessly.
It includes:

Digital Signs Display		
Real-time travel times: Provide current estimates of how long it will take to reach your destination.	**Detours**: Alert drivers to blocked roads and suggests alternative routes to avoid delays and disruptions.	**Alternative routes:** Offer drivers different options to reach their destination.

3. **Safety Systems**
   The goal is to prevent accidents, reduce the severity of crashes, and ensure faster emergency response.



It includes:

Collision avoidance systems: Assist drivers in preventing crashes through automatic braking, steering correction, or pre-warning alerts.

Emergency notification systems: Automatically contact emergency services in case of an accident, improving response time and saving lives.

Driver-assistance features: Provide support to drivers like lane departure warning, blind spot detection, enhancing safety.

4. **Public Transportation Systems (PTS)**
   PTS aims to make public transportation efficient, reliable, and attractive to more users.





It includes:

Integrated ticketing: Allows seamless travel across different transportation modes with a single payment system.

Real-time arrival information: Provides up-to-date information on bus and train arrivals, improving wait times and travel planning.

Smart scheduling of buses and trains: Optimises schedules based on real-time demand and traffic conditions, reducing waiting times and improving service efficiency.

5. **Commercial Vehicle Operations (CVO)**

CVO systems optimise delivery routes, improve fuel efficiency, and reduce emissions.



It includes:

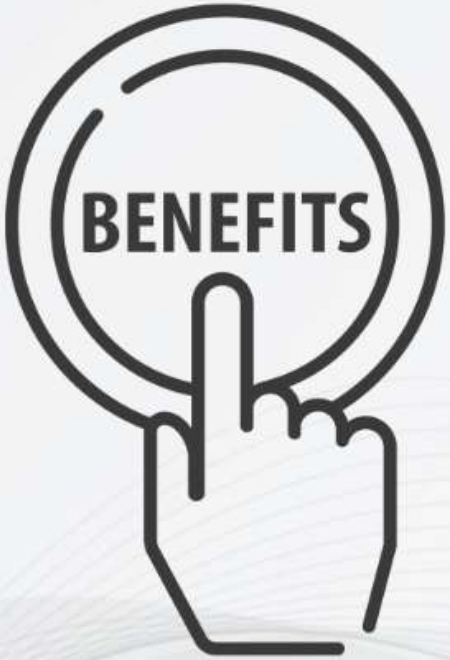
Focuses on tracking: Monitor the location, performance, and health of vehicles in real-time.

Managing fleet operations: Plan routes, dispatch vehicles, and optimise fuel consumption.

Ensuring driver safety: Promote safe driving practices through monitoring, training, and fatigue detection.

Compliance with regulations: Ensure vehicles and drivers meet all legal requirements, including licensing, maintenance, and safety standards.

Benefits of ITS



- Reduced traffic congestion
- Enhanced safety
- Lower emissions
- Increased productivity
- Improved access to transportation

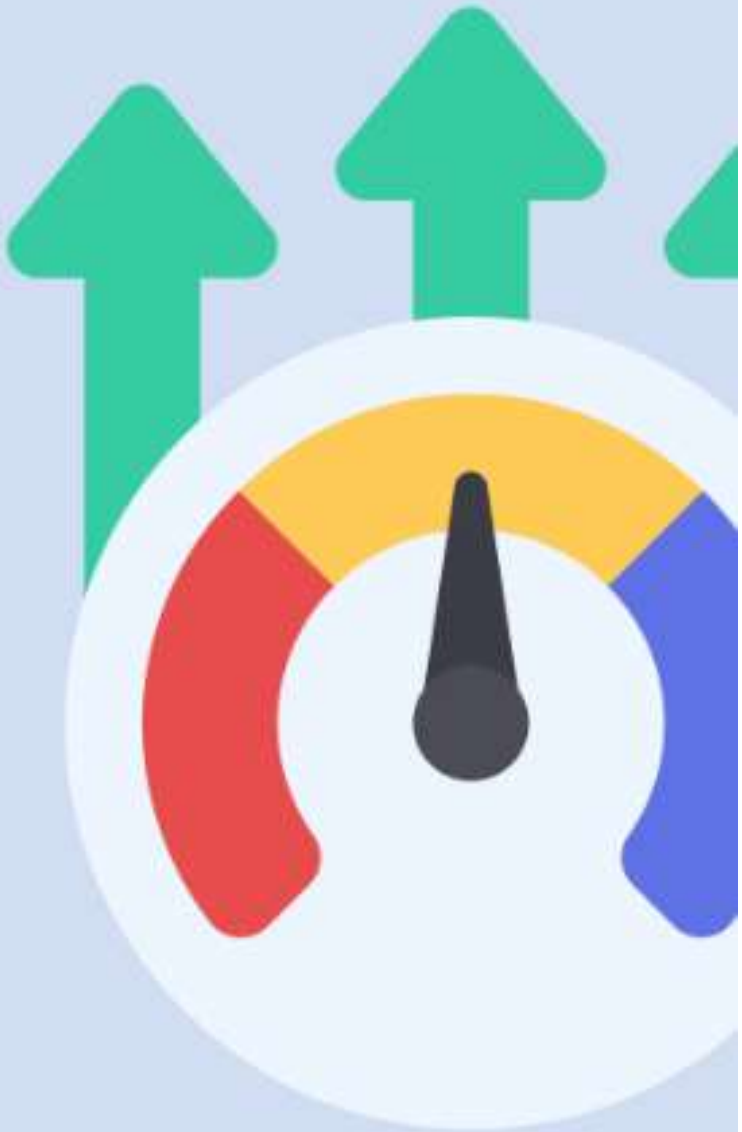
Improved access to transportation:

Includes integrated ticketing, real-time information, and smart scheduling.



Increased productivity:

Less time spent in traffic jams leads to increased work time and productive output.



Lower emissions:

Includes optimised routing, electric vehicles, and smarter traffic management.

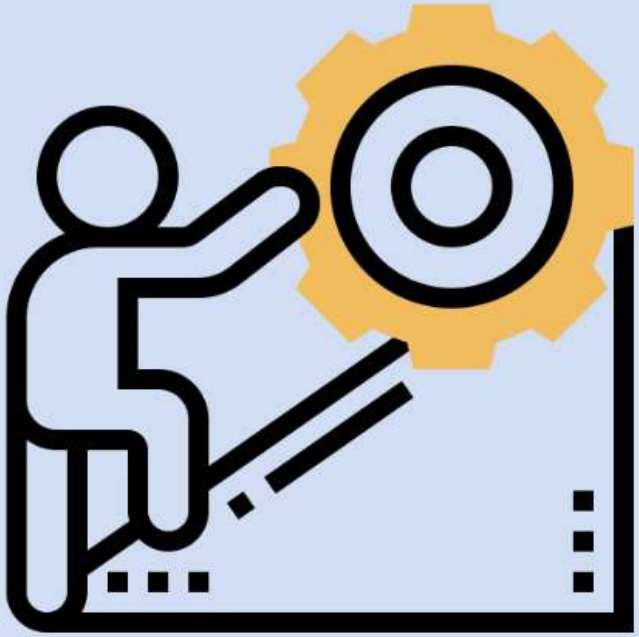


Reduced traffic congestion:

Includes smoother traffic flow through real-time updates, and dynamic lane control.



Challenges of ITS



- High cost of infrastructure
- Data privacy and security
- Equity and access
- Standardisation Interoperability

High cost of infrastructure

High cost of infrastructure

Building and maintaining advanced ITS systems requires significant investment.



High cost of infrastructure

Data privacy and security

Data privacy and security:

Extensive data collection raises concerns about user privacy, potential misuse, and robust security measures.



Equity and access

Standardisation Interoperability

High cost of infrastructure

Data privacy and security

Equity and access

Equity and access:

Not everyone may have equal access to or benefit from ITS technologies, creating potential socio-economic disparities.



Standardisation Interoperability

High cost of infrastructure

Data privacy and security

Equity and access

Standardisation Interoperability

Standardisation and interoperability:

Lack of consistent standards across different systems can hinder seamless data exchange.



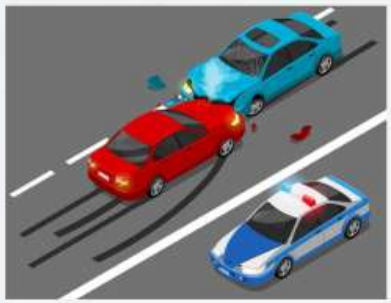
4.1.2: Connecting for Safety

Transportation should not be a gamble.

Transportation safety is **paramount**, and connecting vehicles and infrastructure plays a crucial role.



Connecting for Safety



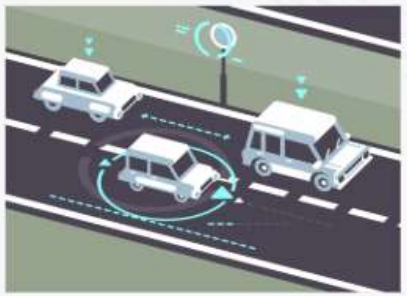
Collision Avoidance Systems

Vehicles communicate with each other and infrastructure to warn drivers of potential hazards and prevent accidents.



Emergency Notification Systems

Cars automatically notify emergency services in case of accidents, reducing response times and saving lives.



Driver-Assistance Systems

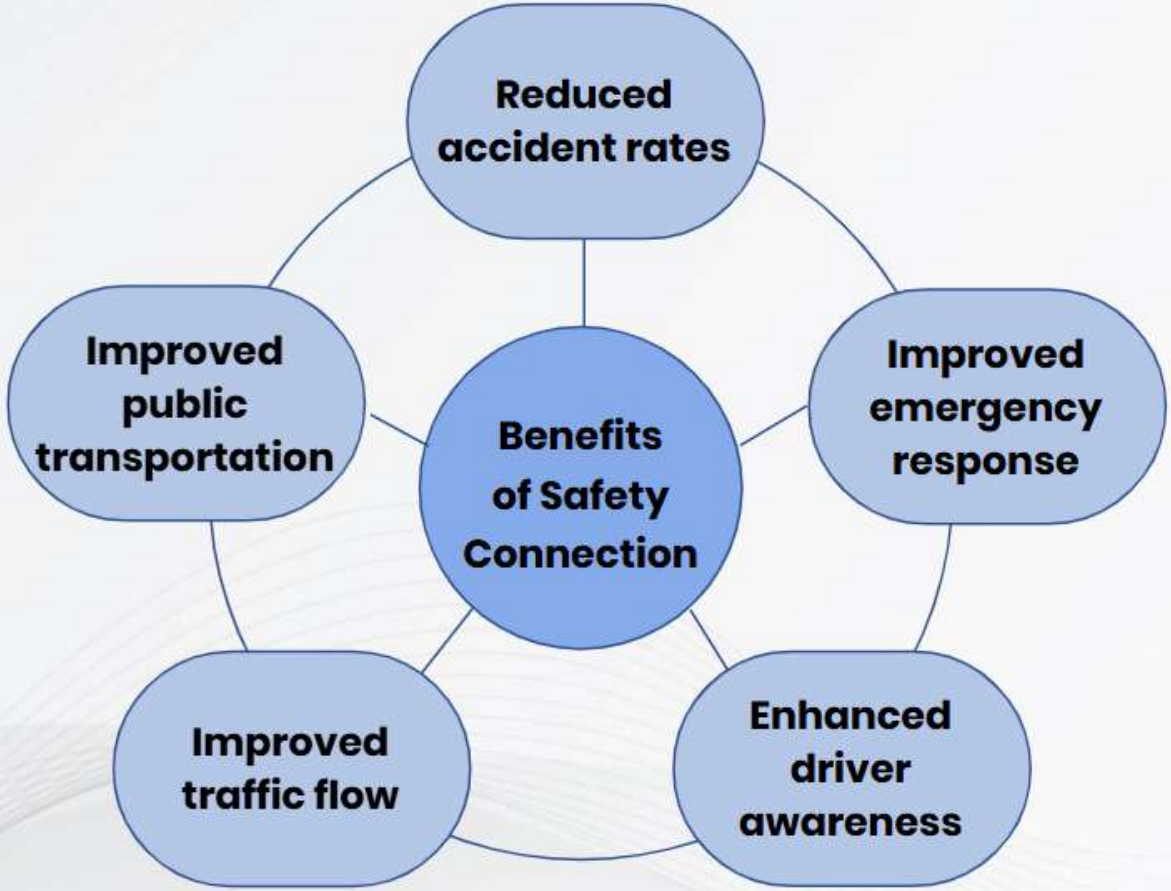
Features like lane departure warning, blind-spot detection, and adaptive cruise control help drivers to avoid collisions.



Advanced Traffic Management

Dynamic signage and road markings warn drivers of hazards and guide them safely through construction zones or weather events.

Benefits of Safety Connection





Future of Safety Connection

Here's how the future of safety connection looks:



- Wider adoption of Vehicle-to-Vehicle technology
- Wider adoption of Vehicle to Infrastructure technology
- Integration with autonomous vehicles
- Personalised safety alerts and Cybersecurity advancements

Wider adoption of Vehicle-to-Vehicle technology

Vehicle-to-Vehicle:

Here cars talk to each other, sharing critical data in real-time to prevent collisions and improve traffic flow.

Wider adoption of Vehicle to Infrastructure technology

Integration with autonomous vehicles

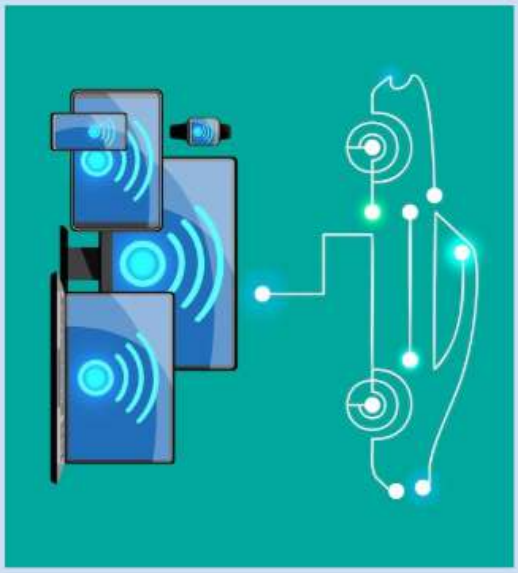
Personalised safety alerts and Cybersecurity advancements

Wider adoption of Vehicle-to-Vehicle technology

Wider adoption of Vehicle to Infrastructure technology

Vehicle-to-Infrastructure:

In this, cars connect to smart infrastructure, receiving updates on hazards, detours, and optimal routes.



Integration with autonomous vehicles

Personalised safety alerts and Cybersecurity advancements

Wider adoption of Vehicle-to-Vehicle technology

Wider adoption of Vehicle to Infrastructure technology

Integration with autonomous vehicles

Autonomous Vehicles:

Includes advanced algorithms enhancing safety and potentially removing human error from driving.



Personalised safety alerts and Cybersecurity advancements

- Wider adoption of Vehicle-to-Vehicle technology
- Wider adoption of Vehicle to Infrastructure technology
- Integration with autonomous vehicles
- Personalised safety alerts and Cybersecurity advancements

Personalised Safety Alerts:

Include warnings based on driver's behaviour, vehicle condition, and environmental factors and also include robust and evolving measures to protect from cyber threats and attacks.



Unit 4.2: Cities of the Future

Key Learning Outcomes



At the end of this unit, you should be able to:

- Discuss the concept of autonomous vehicles.
- Explain emerging technologies and their contributions to future mobility solutions.
- Analyse the ethical considerations of autonomous vehicles.

Cities of the Future

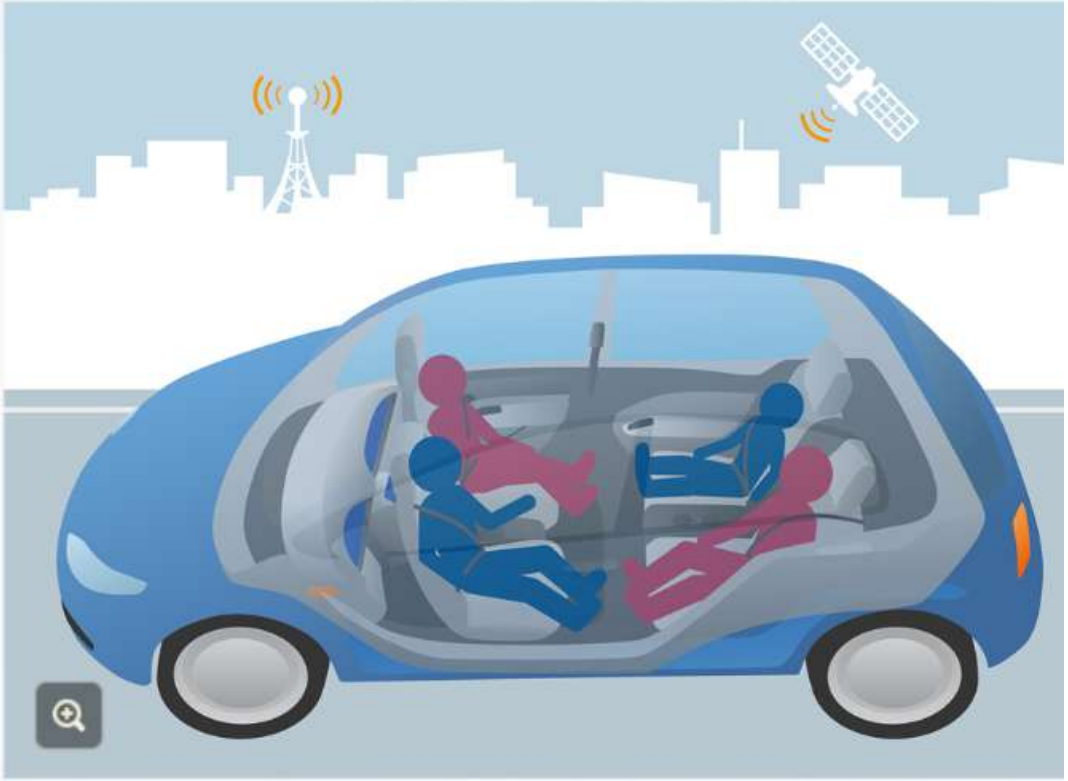
Connected cities and autonomous vehicles are transforming our urban landscapes, building a greener, smarter, and more seamless future for mobility.

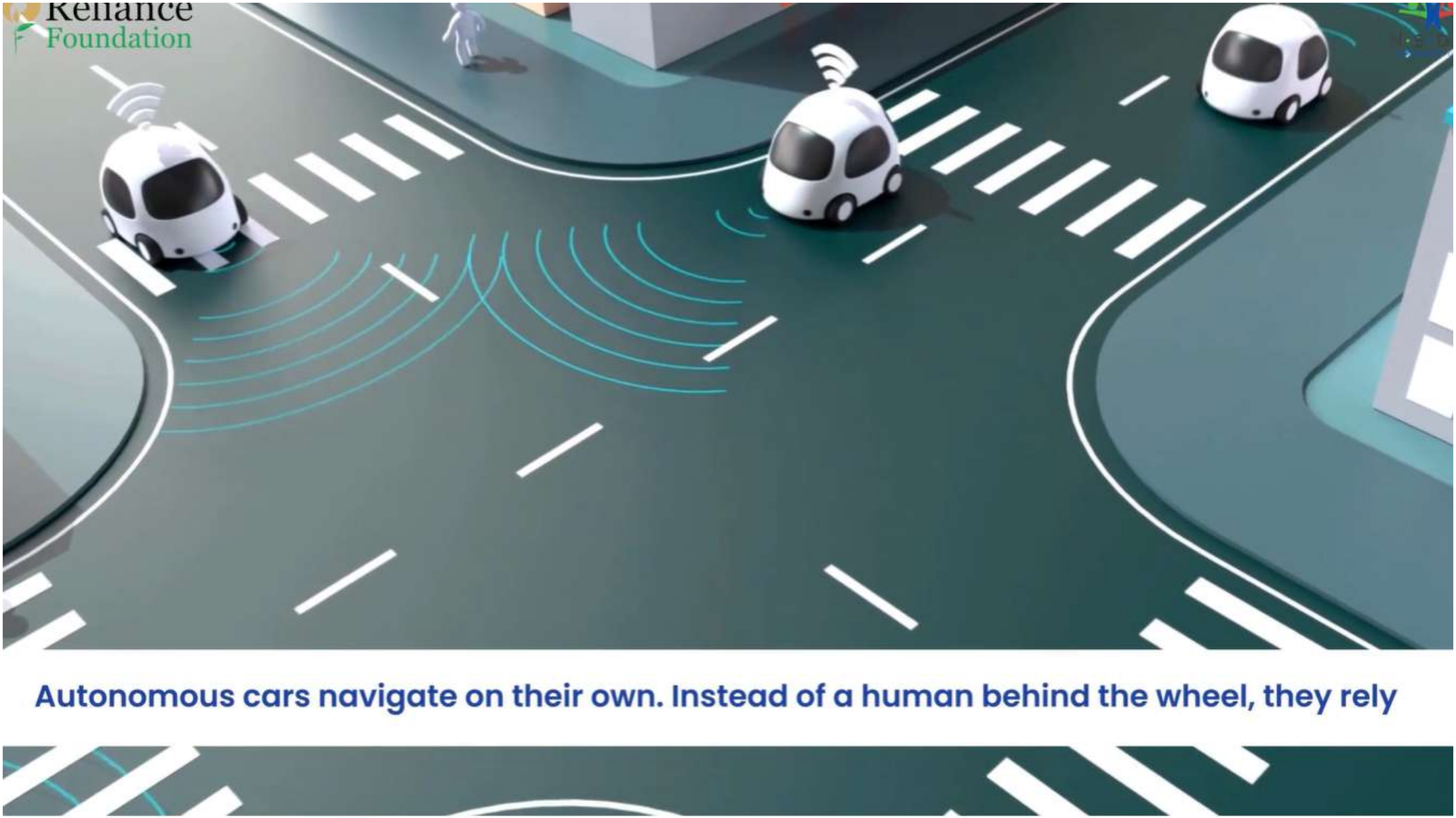


4.2.1: The Rise of Autonomous Vehicles

In the not-so-distant future, the urban landscape may be transformed by a singular innovation which is the rise of autonomous vehicles (AVs).

What are these autonomous vehicles?

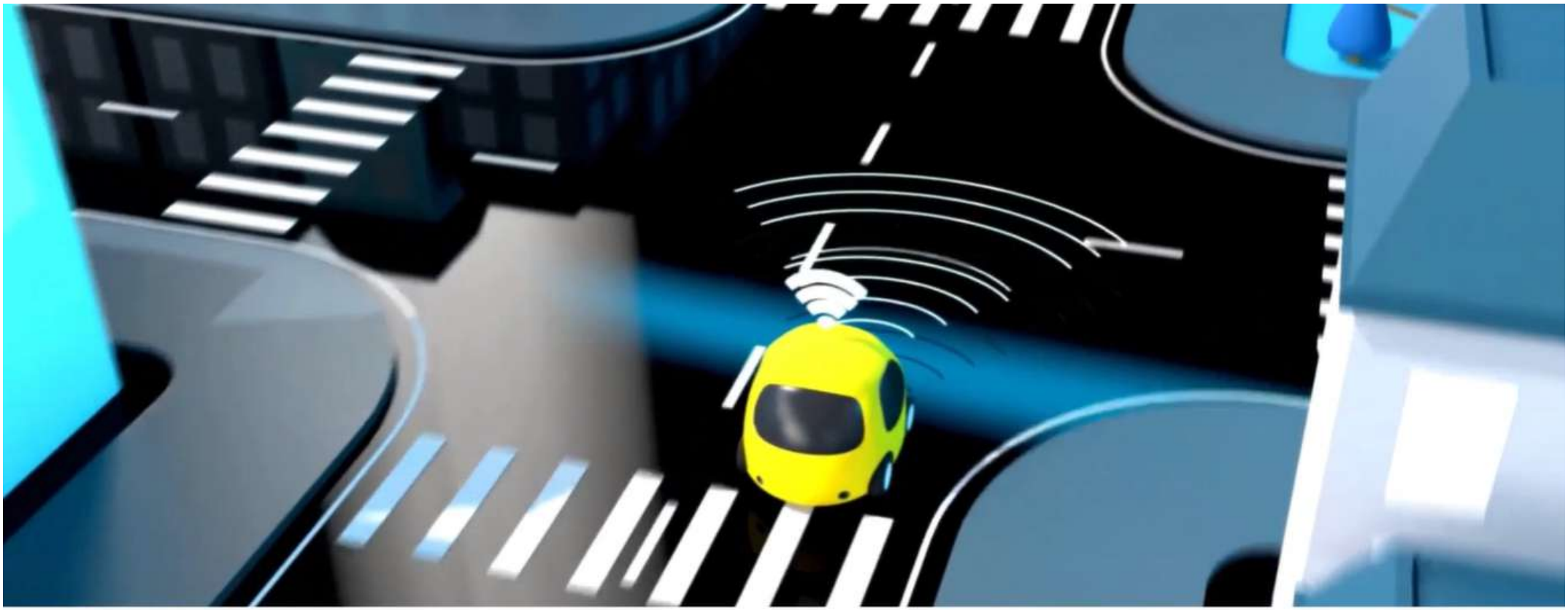




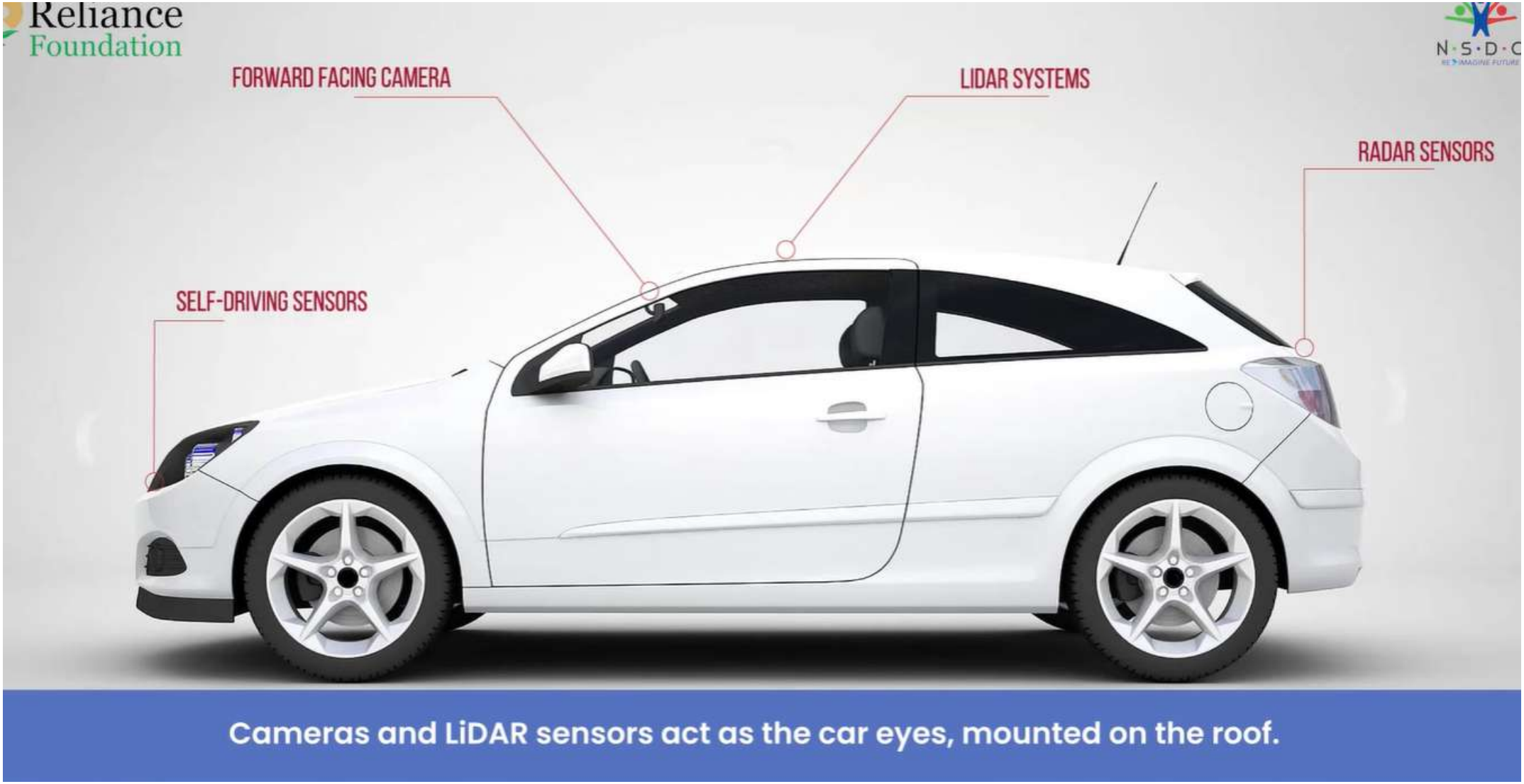
Autonomous cars navigate on their own. Instead of a human behind the wheel, they rely

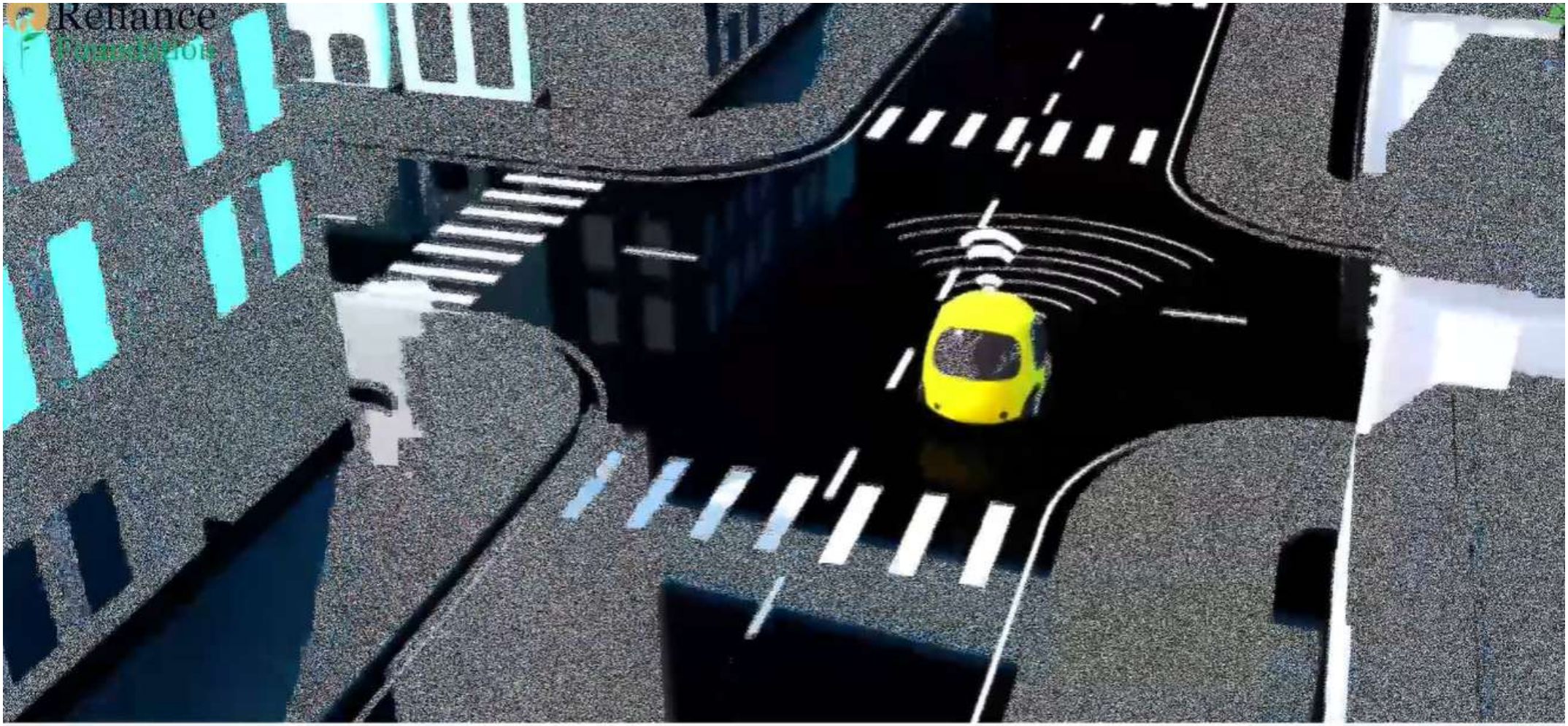


LiDAR sensors, radar, cameras, and ultrasonic sensors - all positioned like watchful eyes across the vehicle.



These sensors become the car's brain, constantly scanning and understanding the world around them.


Reliance
Foundation
N·S·D·C
FORWARD FACING CAMERA
LIDAR SYSTEMS
RADAR SENSORS
SELF-DRIVING SENSORS
Cameras and LiDAR sensors act as the car eyes, mounted on the roof.



Unlike radar's radio waves, it uses light rays that paint a picture of the world with millions of light points. This point helps the car understand its environment in minute detail.



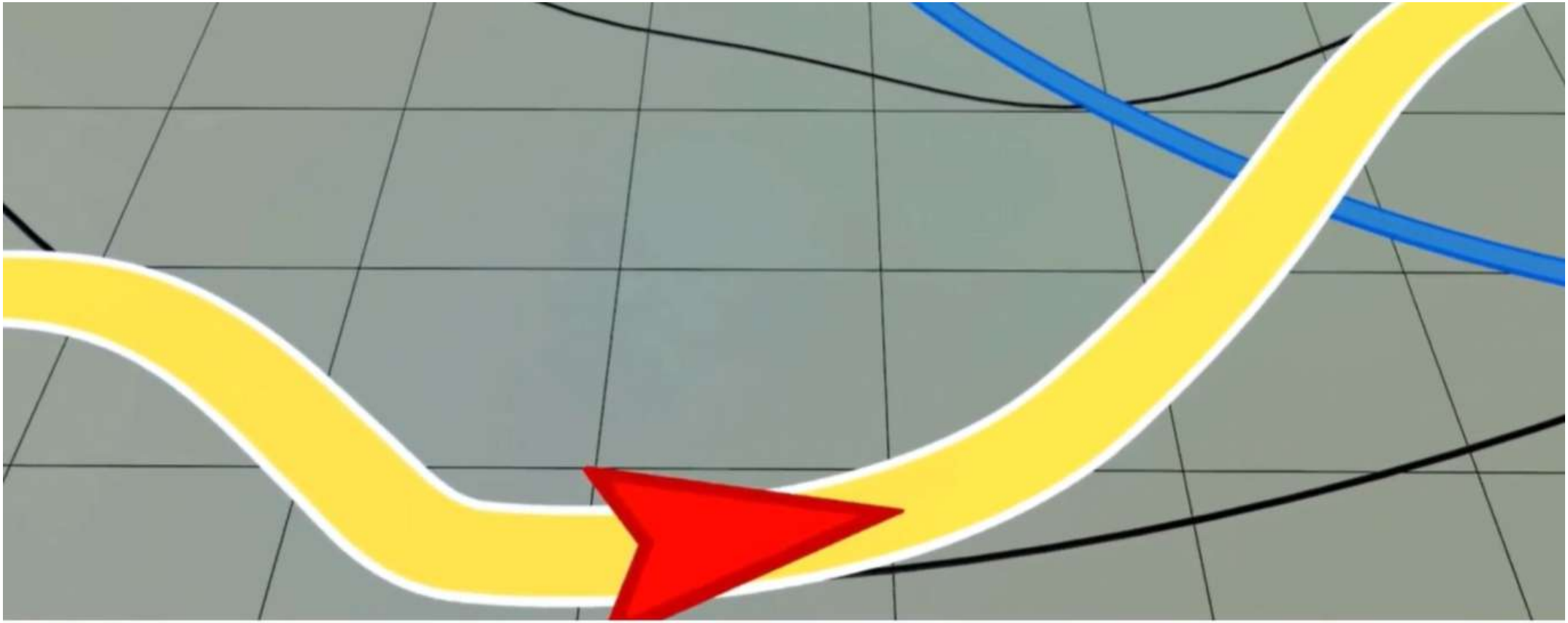
Cameras step in, using artificial intelligence to read and understand street signs. They are also obstacle detectors and lane-keeping assistants, keeping the car on the right track.



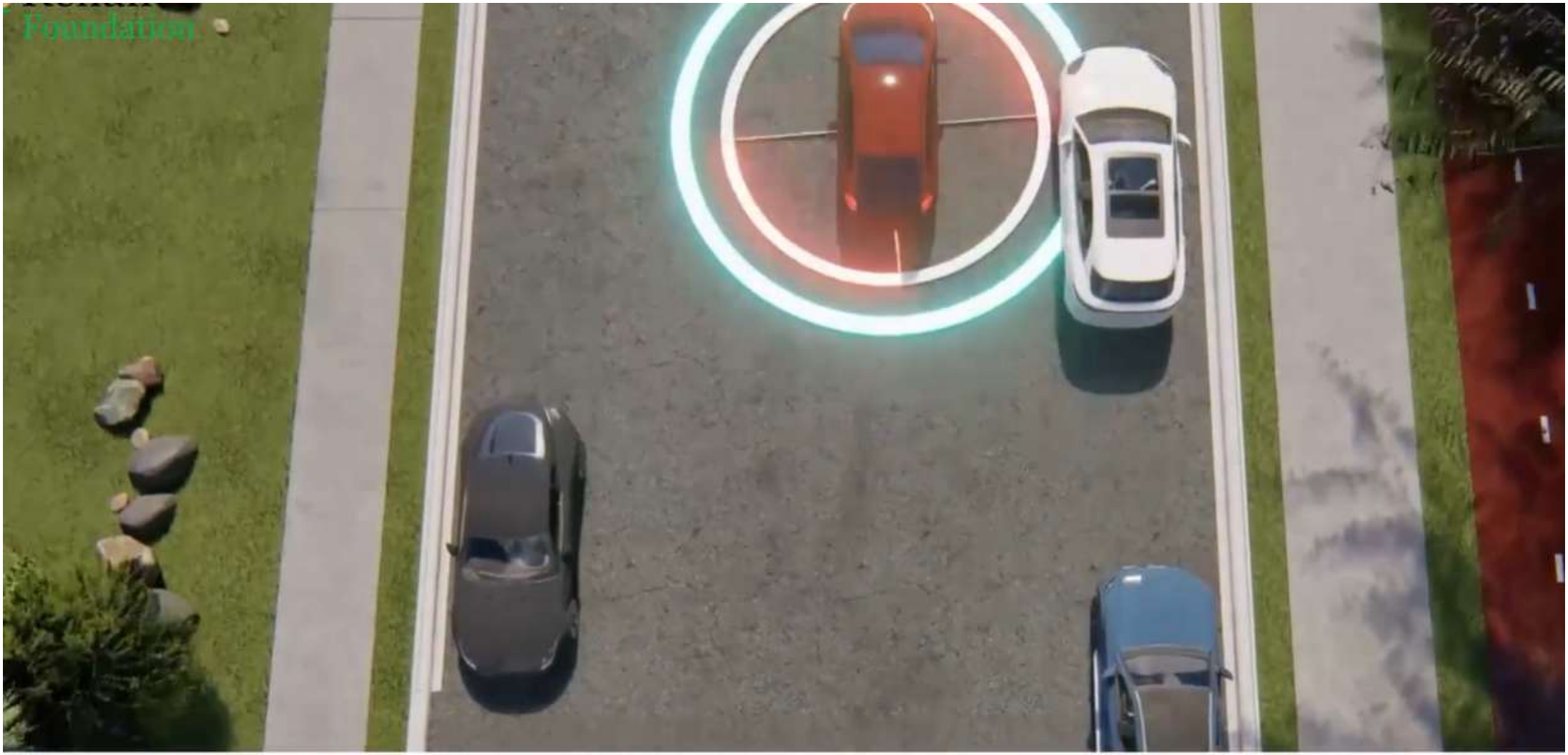
But during fog, both LiDAR and cameras cannot see clearly. That's when radar is used.



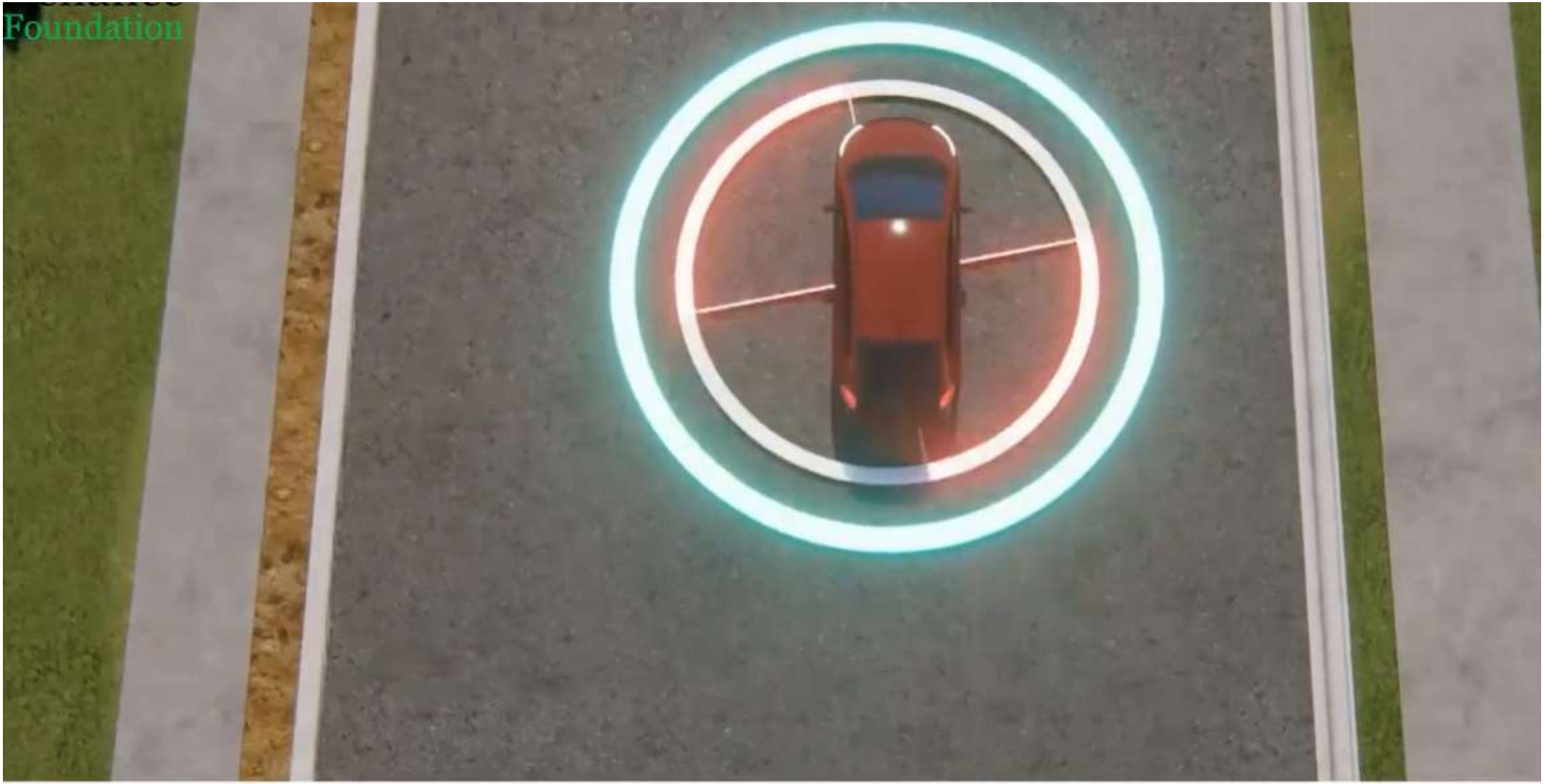
It sends out radio waves to warn the car of potential collisions ahead. But it can only detect objects big enough to reflect its waves.



While radar is great at spotting big things, it might miss pedestrians or cyclists. To know where they are going, these cars use GPS and detailed digital maps with info on speed limits and stop signs.



But GPS isn't perfect, in tunnels or narrow streets. That's when odometry steps in,



using things like wheel diameter and speed to estimate the car position.



Finally, all this data from sensors is combined and processed by powerful comput



to make decisions and keep everyone safe

Types of Autonomy Levels

After understanding the functions of a self-driving car, it is also important to understand the types of autonomy levels a self-driving car has.

There are 6 Autonomous levels in a self-driving car.

Autonomous Levels



Autonomous Levels

Level 0 No Automation (Driver retains full control.)

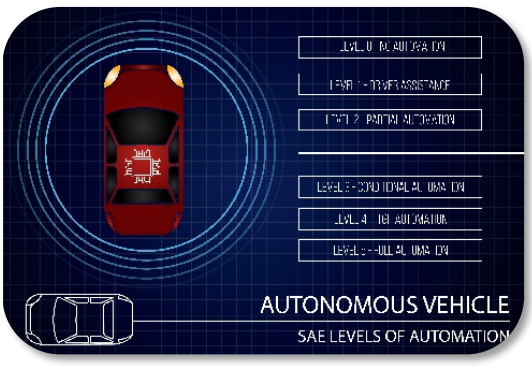
Level 1 Driver Assistance (Feature like lane departure warning assists the driver but do not control the vehicle.)

Level 2 Partial Automation (Systems can control both steering and acceleration in specific situations, like highway driving.)

Level 3 Conditional Automation (Driver can take back control when prompted.)

Level 4 High Automation (AV operates driverless in designated specific scenarios, but human intervention is needed in some situations.)

Level 5 Full Automation (AV operates completely autonomously in all situations.)



Benefits of Autonomous Vehicles

With these various features we can imagine how beneficial autonomous vehicles are. Let us discuss some.

Autonomous Vehicles...

- Reduce traffic accidents caused by human error.
- Advanced algorithms can optimise traffic flow and synchronise vehicles.
- AVs can provide independent mobility for people with disabilities.
- AVs can significantly reduce carbon emissions.

Challenges of AVs:

Some of the major challenges of AVs are:

Technological Hurdles	Regulatory Frameworks	Ethical Considerations	Public Acceptance
Includes sensor reliability, complex software development, and handling unexpected situations.	Robust regulations and safety standards need to be established.	There are questions about job displacement, data privacy, and responsibility in case of accidents.	Building public trust and overcoming concerns about safety is not easy.

4.2.2: Emerging Technologies and Future Mobility

Emerging technologies like drone delivery and **hyperloops** are poised to reshape the future of mobility, nurturing seamless, efficient, and sustainable movement across urban and rural landscapes.

Emerging Technology



Emerging Technologies

Here are some emerging technologies in autonomous vehicle category:

1. Drone Delivery



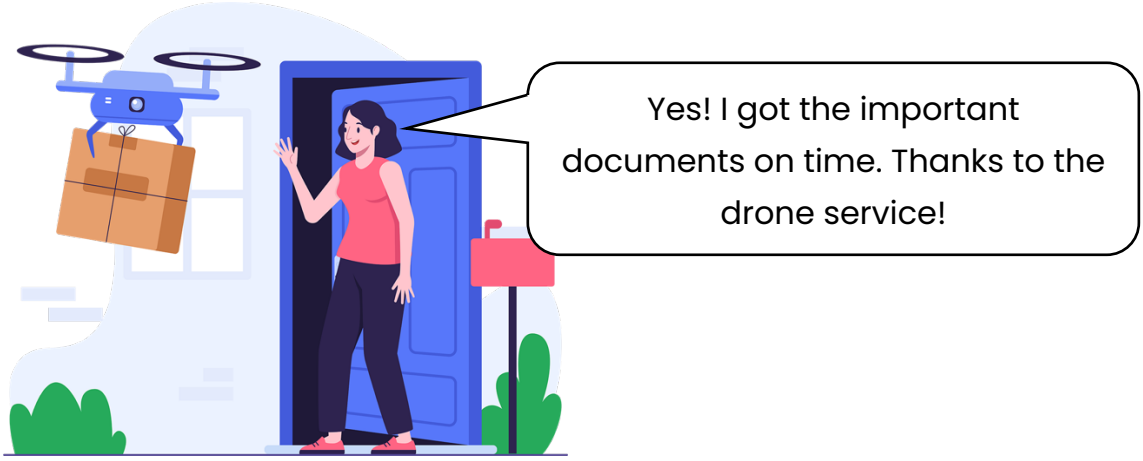
Characteristics:

- Last-mile revolution: Delivering goods directly, bypassing traffic and congestion.
- Rapid response times: Medical supplies and emergency essentials reach destinations in minutes.
- Accessibility in remote areas: Connecting isolated communities with essential services.

Challenges:

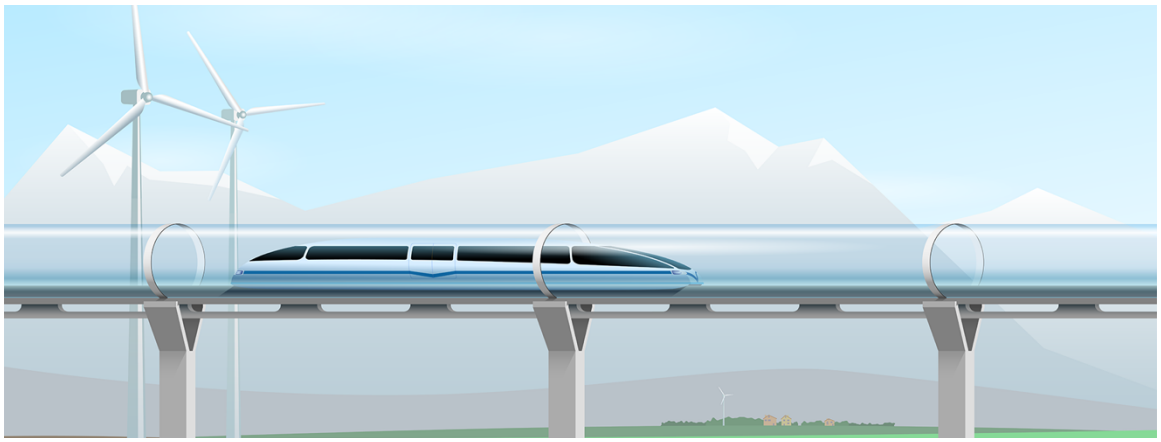
- Regulation and safety: Ensuring safe integration with airspace and mitigating noise concerns.
- Range and payload limitations: Expanding drone capabilities for heavier packages and longer distances.
- Privacy and security: Addressing data collection and potential misuse during deliveries.

Observe this scenario.



So, what did you conclude from this scenario?
Anything can be at your doorstep in the shortest possible time.

2. Hyperloop



Characteristics:

- Ultra-fast travel: Connecting cities in minutes, blurring geographical boundaries.
- Sustainable and efficient: Powered by renewable energy and minimising environmental impact.
- Reduced congestion: Taking pressure off existing transportation networks.

Challenges:

- High construction costs: Building extensive vacuum tube infrastructure requires significant investment.
- Technological hurdles: Maintaining stable pressure and overcoming technical challenges at high speeds.
- Public acceptance: Addressing safety concerns and integrating smoothly with existing transportation systems.

Now let us observe a scenario.



So, what did you conclude from this scenario?

Imagine the amount of time and energy we could save with this technology.